

Program Structure



General:

Live Teaching - Live Teaching sessions from Monday – Saturday from 08:45 AM to 05:45 PM.

For the Graded Assessments (quiz and project), the submission timeline will be

Graded Quiz- 30 minutes

Graded Project- 3 hours

Day No	Date	Day	Course	Topics covered in the day	Sub-Topics	Lab Topic	Description of the Lab	Nuvepro (Yes/No)	Tool Name	Remarks Notes
Day 1	12 January 2026	Monday	Hardware & Networking Basics	Introduction to Program Operating Systems Linux Fundamentals	Operating Systems: Foundation of Operating Systems Windows 10 Mac OS X Android OS Golden Images Linux Fundamentals: Linux Overview Linux Distributions Components of Linux Systems Linux File Systems Bash & Powershell Scripting Boot Loaders Kernels Daemons	Virtual lab Setup on your machine	Students to setup their virtual lab using Virtual Box. They will download the required virtual machines. They will setup the virtual network as per the instructions. Students will download various machines and access them	No	Kali Linux	
Day 2	13 January 2026	Tuesday		Windows Server Fundamentals Computer Hardware	Windows Server Fundamentals Network Basics Defining Server roles Hardware Virtual Server Servers Information: Definition and Purpose) Active Directory DNS Server DHCP Server Web Server FTP Server SSH Server Computer Hardware What is computer memory? Cache memory Common memory errors	Windows Server Administration	Students will create hard disk partitions, setup services in Windows Server 2022. Students will use msinfo32 utility and check the results from their systems. Students will analyze the given msinfo32 output and answer the questions based on the same	No	Windows Server	
Day 3	14 January 2026	Wednesday	Linux Administration & Data Management	Virtualisation & Containerization Basics of Databases	Virtualization Introduction to Virtualization Hypervisor Types of Virtualization Working of a Virtual Machine Benefits of Virtualization Disadvantages of Virtualization Containerization Introduction to Containers Advantages of Containerization Containerization and Security Virtualization vs Containerization Container Technologies Kubernetes Disadvantages of Containerization	Database management lab	Students will install mysql database on their Kali linux machine	No	Kali Linux	
Day 4	15 January 2026	Thursday		Networking Basics IP Addressing	Networking Basics OSI TC/IP Model Different Network Architecture Wireshark: Definition and Purpose IP Addressing MAC Address Introduction to IP Addressing IPv4 Addressing & IPv6 Network Addressing Classless Inter-Domain Routing Private IP Address & Public IP APIPA IPv6 Addressing Network Address Translation Subnet and Super net What is VPN What is a Proxy & Reverse Proxy What is a Load Balancer What is a CDN	Configure IP address in windows machine and Kali linux machine. Practice networking commands	Students will configure IP address in Windows machine and in Kali linux. Ensure that they are in the same network and also ping each other. Students will practice different networking commands. They will build a network of machines using the concept of docker and use different commands. Students will use the DNS protocol to understand the various CDN providers used by different websites Students will access different websites and practice the DNS protocol Students will connect to an open FTP server and upload / download files Students will change the NTP server IP address on their machine	No	Linux ISO & Windows	
Day 5	16 January 2026	Friday	CIA Triad & Cyber Space Risk Management	Network Security & services Network Segmentation	Network Topologies Different types of Devices Types of ports, types of connections Understanding a network diagram DNS NTP HTTP Using Wireshark IP Addressing IPv4 vs IPv6 Network Services Active Directory DNS Email (Exchange Servers) File servers Backups NTP VPN Zero Trust Architecture Network Segmentation Network Architecture Network Monitoring	Building a virtual lab with Cisco Packet tracer Working with different protocols DNS FTP NTP	Create a virtual network with Packet tracer Ping machines across networks Capture packets with Wireahrk and ensure that you get the password Configure NAT Policy (Source NAT) LDAP Integration (Authenticate users with AD Creds) Create a firewall policy	No	Linux ISO & Windows	
Day 6	17 January 2026	Saturday		CIA Triad & Cyber Space Risk Management	CIA Triad Foundations of Infosec Cybersecurity Risk Management (needs more info) Motives Behind Attacks Need for Security What is security posture AI and Cybersecurity IAAA Risk Management basics Types of Risks Risk responses Risk Register	Risk Management	Project: Students to identify risks in the given scenarios	No	Download Cisco Packet Tracer OSINT Tools	
Day 7	19 January 2026	Monday	Basic Concepts of Cybersecurity	Cryptography & It's Types	Types of Cryptography Evolution of Encryption Algorithms Salt Nonce Digital Certificates Digital Signatures Attacks on Crypto Case Study: Heartbleed Vulnerability	Cyberchef Website Scanner Results	Practicing cryptography with Cyberchef Scan a website with the help of a website scanner from Qualys and submit your analysis report on the same	No	Qualys Scanner/ Lab Cyberchef Cyberchef Website Scanner https://gchq.github.io/CyberChef/ https://www.sslab.s.com/ssltest/	

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Day 8	20 January 2026	Tuesday		Introduction to AI Generative AI Risks related to AI AI and Cybersecurity	Introduction to AI Neural Networks Natural Language Processing Computer Vision What is ChatGPT, Microsoft Copilot and Google Bard Generative AI Supervised vs Unsupervised Learning Discriminative AI vs. Generative AI Brief History of Generative AI A Peek into Generative AI Models Large Language Models (LLMs) Introducing ChatGPT - Hands on How does Model understand Text How does Model understand Text Harms possible from AI AI Lifecycle Benefits of AI in Cybersecurity AI Use cases in Cybersecurity State of the World Adapting Gen-AI Enterprise Best Practices for GenAI	ChatGPT Google Gemini	Students to use ChatGPT for various prompts Asking the AI Models to create a phishing email Students to use Gemini for different prompts Asking the AI Models to create a phishing email	No	ChatGPT Google Gemini	
Day 9	21 January 2026	Wednesday		Attack Vectors	What is an attack Surface How to reduce the attack Surface What is a malware Different types of Malware and differences Viruses Worms Trojans Spyware Wiper Malwares Bots Phishing, vishing, whaling and SMShing	Malware Analysis tools	Students to analyze the given malware analysis link and answer the questions	No		Kali Linux
Day 10	22 January 2026	Thursday	Understanding Cyber Attacks	MITRE ATT&CK Framework Case Study	What is a Hacker Different types of Hackers What is a cracker What is an APT What is security testing What is vulnerability assessment and vulnerability management What is Penetration testing Different types of penetrating testing Process of Penetration testing MITRE ATT&CK Framework Intro to MITRE When the bad news hits Indicators of compromise Indicators of compromise - Ransomware Specific Introduction to MITRE ATT&CK Framework The ATT&CK Matrix Case Study: Pwningchang 2018 Cyber Attacks	MITRE ATT&CK Framework	Students to use the MITRE ATT&CK framework and create the defence strategy against a specific attacker group	No		MITRE ATT&CK
Day 11	23 January 2026	Friday		Advanced Persistent Threat Attack Vectors Most famous hackers of the world	<u>Attack Group:</u> <u>Unit 8200</u> <u>Attack Vectors:</u> <u>Phishing</u> <u>Insider Threat</u> <u>Credential Attacks</u> <u>Most famous hackers of the world</u> <u>Kevin Mitnick</u> <u>Julian Assange</u> <u>Adrian Lamo</u> <u>Jeff Moss</u> <u>The Anonymous</u> <u>The Lizard Squad</u> <u>Equation Group</u>	Password cracking with Kali linux	Students will use Kali linux to crack a series of passwords using different tools	No	Kali Linux	
Day 12	24 January 2026	Saturday	Graded Quiz & Project 1	Fundamentals						Kali
Day 13	27 January 2026	Tuesday	VIVA 1							
Day 14	28 January 2026	Wednesday		Introduction to Scripting Content Management using Git Foundations of Java language Classes and Inheritance	Writing Basic Scripts Different tools for programming Creating Git Repos Uploading and Managing the information on git Fork and Repo Management Software Setup Language Understanding/Fundamentals importance of Java Data Types Variables Operators Input / output statements Conditional & looping Statements Wrapper Classes Arrays Strings Methods	HandsOn fundamentals of java programming	Students will be hands on coding along with the faculty	No	Not required	
Day 15	29 January 2026	Thursday		OOPs using Java & Core Java	Introduction to OOPs (class, object, inheritance, Constructor, Polymorphism) Exception Handling (Throw, Throws, Trycatch, Finally) Collection Framework (List, Set, Map, LinkedList, ArrayList, HashSet) Concurrency Streams API Date & Time API Functional Interface Lambda Expressions Optional Class	HandsOn - OOPs, Exception Handling Collection	Students will perform hands on problems based on OOPS, Exception Handling and collect	No	Not required	
Day 16	30 January 2026	Friday	Introduction to Programming - Java	Data Management with Java & SQL	Introduction to Databases: Overview of Databases Types of Databases (Relational, NoSQL, Graph, etc.) Importance and Benefits of Databases Database Management Systems (DBMS) Introduction to Structured Query Language (SQL) SQL DDL, DML, and DQL: Data Definition Language (DDL) in SQL Creating and Altering Database Objects (Tables, Views, Indexes, Constraints) Data Manipulation Language (DML) in SQL Inserting, Updating, Deleting Data in Tables Data Query Language (DQL) in SQL Retrieving Data from Tables (SELECT Statements, Filtering, Sorting, Joining) JDBC: Introduction to JDBC (Java Database Connectivity) JDBC Architecture Establishing Database Connection Executing SQL Queries using JDBC Handling Transactions Handling Exceptions in JDBC Closing Database Resources	Select query & JDBC	Perform CRUD operations using JDBC	No	Not required	
Day 17	31 January 2026	Saturday		Web services	Web Services: Overview of Web Services Types of Web Services (RESTful, SOAP-based, etc.) Advantages and Disadvantages of Web Services Architecture of Web Services Role of XML and JSON in Web Services REST and SOAP: Introduction to RESTful Web Services Introduction to SOAP-based Web Services Comparison between REST and SOAP Characteristics of RESTful Web Services Characteristics of SOAP-based Web Services Consuming REST API / Rest Services:	Hands on REST & SOAP	Students will perform REST & SOAP Functions	No	Not required	

Day 18	02 February 2026	Monday		REST API / Rest Services & Postman	Basics of REST API HTTP Methods (GET, POST, PUT, DELETE) Request and Response Formats (JSON, XML) Authentication and Authorization in REST APIs Error Handling in RESTful Services Asynchronous Operations (Async/Await, Promises) Consuming RESTful Services in various programming languages (Java, Python, etc.) Postman: Introduction to Postman Installing and Setting Up Postman Creating Requests (GET, POST, PUT, DELETE) Handling Request Parameters (Query Parameters, Path Parameters, Headers) Handling Authentication (Basic Auth, Bearer Token, OAuth) Handling Response Data Organizing Requests into Collections Writing Test Scripts in Postman Exporting and Importing Collections	REST APIs and postman	Students will perform hands-on on REST & postman along with faculty.	No	Not required	
Day 19	03 February 2026	Tuesday		Introduction to sailpoint Attacks on Identities	Product Architecture, Installation and Deployment IdentityIQ Preview IdentityCubes Introduction Attacks on credentials Lateral Movements Uber Case Study	Sailpoint Tool Introduction & Key features In sailpoint lab Identity based attacks	Walkthrough into Sailpoint IIQ software Attacking identities on cloud Attacking MFA with burpsuite	Yes	Sailpoint/ Burpsuite	Burpsuite to be installed locally https://www.kali.org/get-kali/#kali-virtual-machines Download Virtual box for images
Day 20	04 February 2026	Wednesday	Identity Management & Identity Governance	Implementing Sailpoint	Identity Cubes Identity Cube Overview Applications/Connectors Identity Mappings Aggregation of Data Capability/Scoping/Workgroups Logs and troubleshoot of build map custom attributes creation mapping correlation manager correlation email id creation unique username creation Application Onboarding, Correlation Onboarding Applications Account Correlation, Manager Correlation Application Configuration Account Reconciliation Ent reconciliation Provisioning Database onboarding Database Directory account Reconciliation Ent reconciliation Provisioning	Understanding Identity vs. Account Reconciliation Identity Cubes	Create and Link: Create several identities and corresponding accounts on different systems (e.g., email, HR system, CRM). Manually link these accounts to a single identity to illustrate the concept of one identity having multiple accounts. Perform Reconciliation: Import account data from multiple systems and reconcile this data with existing identities in the identity repository. Identify and resolve discrepancies (e.g., duplicate accounts, orphaned accounts, and mismatches). Configure Synchronization: Set up a synchronization between a trusted source and multiple target systems, ensuring that when an identity is updated in the trusted source, it automatically updates in the target systems. Construct a Basic Identity Cube: Create a basic Identity Cube for a fictional user by manually aggregating data from multiple sources (e.g., HR system, email, CRM). Include attributes like name, role, access rights, and activity logs. Visualization: Use a tool or drawing to visually represent the Identity Cube, highlighting the aggregation of data from different sources.	Yes	Sailpoint	Sailpoint lab needs to have windows active directory connected to the same server as sailpoint - inform Nuvepro (Same lab continuation from Week 2 + Sailpoint)
Day 21	05 February 2026	Thursday	Graded Quiz & Project 2							GL Platform
Day 22	06 February 2026	Friday	VIVA 2							
Day 23	07 February 2026	Saturday		Implementing Sailpoint	Risk and Policies Joiner : Same day and future onboarding (birthright) Mover : Supervisor change Leaver : disable , disabled OU move : 60 days in disabled ou then delete Provisioning plan : account request, ent request, create/modify/delete/disable/enable/pwd reset rule f) Request based access and approval workflow Certification Certification and Access Reviews Types of Certifications Certification Lifecycle	Loading and Correlating a Financial Application	Onboard a new application by defining its attributes, roles, and access controls within the identity governance system. Create appropriate csv file for financial application Configure account correlation rules to integrate the new application with existing identities. Test the onboarding by creating a new user and provisioning access to the application.	Yes	Sailpoint	
Day 24	09 February 2026	Monday	Identity Management & Identity Governance		Certifications Roles Troubleshooting Debugging common issues in Sailpoint Debug Page Best Practices Reporting Workflow	Troubleshooting and Reporting and Workflow	Debugging common errors Detect misconfiguration Workflow troubleshooting	Yes	Sailpoint	
Day 25	10 February 2026	Tuesday		Reporting and Monitoring using Sailpoint	Analyzing the given data for identifying issues Lifecycle Manager Workflows in Identity IQ	Workflow, LCM introduction	The Lifecycle Manager in SailPoint IdentityIQ automates the process of managing user identities across their lifecycle, from creation through to modification and eventual deletion. Failed Operations: Identify any operations that frequently fail, such as password resets or access requests. Determine if these failures are due to system misconfigurations, insufficient access rights, or external system issues. Compliance Violations: Check for any instances where lifecycle events may have led to compliance violations, such as not removing access for terminated employees promptly Practice and develop skill on SailPoint Workflow manager	Yes	Sailpoint	
Day 26	11 February 2026	Wednesday		Introduction to CyberArk & It's features	Microsoft Active Directory Definition and history of Active Directory. The role of Active Directory in IDAM.Domain Controllers (DCs); Purpose and function. Organizational Units (OUs), Domains, and Forests: Structure and hierarchy. Installing and configuring Domain Controllers. Managing user and computer accounts. Implementing Group Policy Objects (GPOs) for policy management. Synchronization with Azure AD for cloud-based resources.(Not required) Integration with third-party IDAM solutions.(Not Required) Features: Port awareness Vault installation AD integration Email notification	CyberArk Installation	Setting up CyberArk Environment	No	Theoretical	

Day 27	12 February 2026	Thursday	Introduction to CyberArk	CyberArk	Enterprise level identity basics Learn about enterprise identities risks from unmanaged enterprise identities Creating a safe with CyberArk Managing a password safe LDAP User addition Dual Control Object Level Access Control Platform Management Platform creation Platform configuration Platform import/export Master Policy Master Policy creation Platform Performance Tuning Reporting and Monitoring PVWA Reports PrivateArk Client reports Configuring various reports Monitoring the CyberArk Alerts Creating SOPs for the common alerts(Not Required)	Platform Creation	Master Policy & Dual Control Master Policy creation Platform Performance Tuning	No	Theoretical								
Day 28	13 February 2026	Friday			PAM						ManageEngine PAM360	PAM Basics with reference to ManageEngine PAM360 PAM360 Installation scenarios <u>Terminology, Modules and Features</u> PAM360 Architecture PAM360 Components Databases supported User and User groups, <u>Adding users to PAM360</u> Authentication options in PAM360 Active Directory, LDAP Resources an Resource Groups, Managing Resources and Accounts Managing Accounts and Passwords Zero Trust Approach in PAM 360	PAM360 installation	Students to install PAM360 software on Windows VM	Yes	ManageEngine PAM360	
Day 29	14 February 2026	Saturday										Adding users to PAM360	Adding users to PAM360 manually	Yes	ManageEngine PAM360		
Day 30	16 February 2026	Monday	Managing Accounts and Passwords in PAM360	Adding and Managing Accounts and Passwords for users in PAM360		Yes	ManageEngine PAM360										
Day 31	17 February 2026	Tuesday	Cloud Based Identity Management	Microsoft Azure and Identity Management	Introduction to various Identity services in Azure	IAM in Azure	Configure and test IAM settings in Microsoft Azure Explore identity services	Yes	Azure								
Day 32	18 February 2026	Wednesday			Single Sign on Different Types of Authentication mechanisms	Configuring different types of authentication	Configuring different types of authentication	Yes	Azure								
Day 33	19 February 2026	Thursday		Microsoft Azure and Identity Management	SAML OAuth OIDC MFA Cookies and Sessions HTTP error codes <u>Proxy and Reverse proxy</u> Integrate SAML and OAuth applications with Azure SSO	Configuring and Debugging the authentication	Configure different session cookie settings Analyzing cookies Configuring proxy with Burpsuite and modify the request as it goes to the server	Yes	Azure								
Day 34	20 February 2026	Friday	SAML and OAuth in Azure		Implement SSO in Azure with the help of SAML and OAuth	Yes	Azure										
Day 35	21 February 2026	Saturday	VIVA 3														
Day 36	23 February 2026	Monday	Graded Quiz & Project 3		Manage Engine PAM 360 & Sailpoint				PAM 360 VM								